

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU	DRUMS & DESALTER DATASHEET ASME VIII Div.1 - pressure vessels D-101A - Electrostatic desalter, 1st stage	DOC. No.	CFU-000-ME-DS-002	
		REV / DATE	A / 2026-10-02	
		STATUS	Issued for review (FEED)	
		PREPARED	CLAUDE CODE	CHK: - APP: -
		SHEET	1	

GENERAL					
1	Tag / quantity	D-101A / 1	2	Service	Electrostatic desalter, 1st stage
3	Orientation / support	Horizontal, 2 saddles	4	Code	ASME VIII Div.1 (U-stamp)
5	Size (ID x T/T)	ID 3.8 m x 30.0 m T/T, AC/DC dual-polarity grids	6	Volume	354.6 m3

DESIGN CONDITIONS					
7	Operating P / T	11.5 barg / 136 C	8	Design P / T	17.0 barg / 165 C
9	External design P	None (see note)	10	MDMT	-5 C
11	Corrosion allowance	3.0 mm	12	Joint efficiency	1.0
13	Hydrotest pressure	22.10 barg	14	Hold-up / residence	27.7 min
15	PWHT	No (code)	16	Sour service	-

MECHANICAL					
17	Material shell / heads	SA-516-70	18	Shell t (req / nominal)	27.0 / 28 mm
19	Head t 2:1 SE (req / nominal)	26.9 / 28 mm	20	Boot	-
21	Bottom of shell elevation	EL 101.500	22	Insulation	50 mm
23	Internals	Electrostatic grids (AC/DC), inlet distributor, outlet ~	24	Electrical (desalter)	3 x transformers, 100 % reactance, 480 V / 16.5-23 ~

WEIGHTS					
25	Fabricated	119.0 t	26	Empty	137.4 t
27	Operating	414.0 t	28	Hydrotest	487.1 t

NOZZLE SCHEDULE (SEE GA CFU-100-ME-GA-007)						
Mark	Qty	Size	Rating	Face	Service	Location
N1	1	14"	300#	RF	Crude + wash water inlet (to distributor header)	15.0 m from LH TL
N2	1	16"	300#	RF	Desalted crude outlet (collector header)	15.0 m from LH TL
N3A/B	2	4"	300#	RF	Brine outlet (to E-118 / WWT)	7.5 m from LH TL
N4A/B	2	3"	300#	RF	Mud-wash inlet (from P-118)	4.5 m from LH TL
N5A-D	4	4"	300#	RF	Sludge drain / mud-wash outlet	3.0 m from LH TL
N6	1	4"	300#	RF	PSV (PSV-1002/1003, 4M6)	25.5 m from LH TL
N7	1	2"	300#	RF	Vent / N2 purge	27.6 m from LH TL
N8A-C	3	8"	300#	RF	Transformer entrance bushing	9.9 m from LH TL
N9A/B	2	2"	300#	RF	Interface level (LT / LDT) - capacitance / RF	18.0 m from LH TL
N10A-E	5	1"	300#	RF	Try-cocks / interface sample	20.4 m from LH TL
N11	1	3"	300#	RF	Drain	22.5 m from LH TL
N12	1	2"	300#	RF	Steam-out	2.4 m from LH TL
N13	1	2"	300#	RF	Low-level / low-low level switch (LSLL - transformer trip)	16.5 m from LH TL
M1/M2	2	24"	300#	RF	Manway 24" (one per head)	0.0 m from LH TL

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		REV / DATE	A / 2026-10-02	
		STATUS	Issued for review (FEED)	
		PREPARED	CLAUDE CODE	CHK: - APP: -
		SHEET	2	

GENERAL					
1	Tag / quantity	D-101B / 1	2	Service	Electrostatic desalter, 2nd stage
3	Orientation / support	Horizontal, 2 saddles	4	Code	ASME VIII Div.1 (U-stamp)
5	Size (ID x T/T)	ID 3.8 m x 30.0 m T/T, AC/DC dual-polarity grids	6	Volume	354.6 m3

DESIGN CONDITIONS					
7	Operating P / T	11.5 barg / 136 C	8	Design P / T	17.0 barg / 165 C
9	External design P	None (see note)	10	MDMT	-5 C
11	Corrosion allowance	3.0 mm	12	Joint efficiency	1.0
13	Hydrotest pressure	22.10 barg	14	Hold-up / residence	27.7 min
15	PWHT	No (code)	16	Sour service	-

MECHANICAL					
17	Material shell / heads	SA-516-70	18	Shell t (req / nominal)	27.0 / 28 mm
19	Head t 2:1 SE (req / nominal)	26.9 / 28 mm	20	Boot	-
21	Bottom of shell elevation	EL 101.500	22	Insulation	50 mm
23	Internals	Electrostatic grids (AC/DC), inlet distributor, outlet ~	24	Electrical (desalter)	3 x transformers, 100 % reactance, 480 V / 16.5-23 ~

WEIGHTS					
25	Fabricated	119.0 t	26	Empty	137.4 t
27	Operating	414.0 t	28	Hydrotest	487.1 t

NOZZLE SCHEDULE (SEE GA CFU-100-ME-GA-007)						
Mark	Qty	Size	Rating	Face	Service	Location
N1	1	14"	300#	RF	Crude + wash water inlet (to distributor header)	15.0 m from LH TL
N2	1	16"	300#	RF	Desalted crude outlet (collector header)	15.0 m from LH TL
N3A/B	2	4"	300#	RF	Brine outlet (to E-118 / WWT)	7.5 m from LH TL
N4A/B	2	3"	300#	RF	Mud-wash inlet (from P-118)	4.5 m from LH TL
N5A-D	4	4"	300#	RF	Sludge drain / mud-wash outlet	3.0 m from LH TL
N6	1	4"	300#	RF	PSV (PSV-1002/1003, 4M6)	25.5 m from LH TL
N7	1	2"	300#	RF	Vent / N2 purge	27.6 m from LH TL
N8A-C	3	8"	300#	RF	Transformer entrance bushing	9.9 m from LH TL
N9A/B	2	2"	300#	RF	Interface level (LT / LDT) - capacitance / RF	18.0 m from LH TL
N10A-E	5	1"	300#	RF	Try-cocks / interface sample	20.4 m from LH TL
N11	1	3"	300#	RF	Drain	22.5 m from LH TL
N12	1	2"	300#	RF	Steam-out	2.4 m from LH TL
N13	1	2"	300#	RF	Low-level / low-low level switch (LSLL - transformer trip)	16.5 m from LH TL
M1/M2	2	24"	300#	RF	Manway 24" (one per head)	0.0 m from LH TL

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		DRUMS & DESALTER DATASHEET ASME VIII Div.1 - pressure vessels D-102 - Atm. overhead reflux drum (3-phase)		DOC. No. CFU-000-ME-DS-002 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 3	
GENERAL					
1	Tag / quantity	D-102 / 1	2	Service	Atm. overhead reflux drum (3-phase)
3	Orientation / support	Horizontal, 2 saddles	4	Code	ASME VIII Div.1 (U-stamp)
5	Size (ID x T/T)	ID 3.4 m x 12.0 m T/T; boot ID 1.5 m x 3.0 m	6	Volume	124.5 m3
DESIGN CONDITIONS					
7	Operating P / T	0.7 barg / 45 C	8	Design P / T	3.5 barg / 160 C
9	External design P	None (see note)	10	MDMT	-5 C
11	Corrosion allowance	6.0 mm	12	Joint efficiency	1.0
13	Hydrotest pressure	4.55 barg	14	Hold-up / residence	10 min
15	PWHT	Yes (HIC-resistant / sour)	16	Sour service	NACE MR0103
MECHANICAL					
17	Material shell / heads	SA-516-70 (HIC-tested)	18	Shell t (req / nominal)	11.9 / 12 mm
19	Head t 2:1 SE (req / nominal)	11.9 / 12 mm	20	Boot	ID 1.5 m x 3.0 m
21	Bottom of shell elevation	EL 107.000	22	Insulation	None
23	Internals	Inlet diverter, vortex breakers, weir / boot	24	Electrical (desalter)	-
WEIGHTS					
25	Fabricated	19.4 t	26	Empty	19.4 t
27	Operating	63.0 t	28	Hydrotest	144.0 t
NOZZLES					
Nozzle sizes per P&ID line sizing (CFU-100/200-PR-PID); manway 24" on head; PSV, vent 2", drain 2", steam-out 2", LT/LG bridle 2".					

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GENERAL					
1	Tag / quantity	D-103 / 1	2	Service	Fuel gas knock-out drum
3	Orientation / support	Vertical, legs	4	Code	ASME VIII Div.1 (U-stamp)
5	Size (ID x T/T)	ID 0.9 m x 2.5 m T/T	6	Volume	1.8 m3
DESIGN CONDITIONS					
7	Operating P / T	3.5 barg / 30 C	8	Design P / T	7.0 barg / 90 C
9	External design P	None (see note)	10	MDMT	-5 C
11	Corrosion allowance	3.0 mm	12	Joint efficiency	0.85
13	Hydrotest pressure	9.10 barg	14	Hold-up / residence	20 min
15	PWHT	No (code)	16	Sour service	-
MECHANICAL					
17	Material shell / heads	SA-516-70	18	Shell t (req / nominal)	6.4 / 8 mm
19	Head t 2:1 SE (req / nominal)	6.4 / 8 mm	20	Boot	-
21	Bottom of shell elevation	EL 101.000	22	Insulation	None
23	Internals	Inlet diverter, vortex breakers	24	Electrical (desalter)	-
WEIGHTS					
25	Fabricated	1.2 t	26	Empty	1.2 t
27	Operating	1.8 t	28	Hydrotest	3.0 t
NOZZLES					
Nozzle sizes per P&ID line sizing (CFU-100/200-PR-PID); manway 24" on head; PSV, vent 2", drain 2", steam-out 2", LT/LG bridle 2".					

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GENERAL					
1	Tag / quantity	D-104 / 1	2	Service	Flare knock-out drum (unit)
3	Orientation / support	Horizontal, 2 saddles	4	Code	ASME VIII Div.1 (U-stamp)
5	Size (ID x T/T)	ID 2.1 m x 6.5 m T/T	6	Volume	24.9 m3
DESIGN CONDITIONS					
7	Operating P / T	0.2 barg / 60 C	8	Design P / T	3.5 barg / 280 C
9	External design P	None (see note)	10	MDMT	-5 C
11	Corrosion allowance	3.0 mm	12	Joint efficiency	1.0
13	Hydrotest pressure	4.60 barg	14	Hold-up / residence	20 min
15	PWHT	No (code)	16	Sour service	-
MECHANICAL					
17	Material shell / heads	SA-516-70	18	Shell t (req / nominal)	7.6 / 8 mm
19	Head t 2:1 SE (req / nominal)	7.6 / 8 mm	20	Boot	-
21	Bottom of shell elevation	EL 101.000	22	Insulation	50 mm
23	Internals	Inlet diverter, vortex breakers	24	Electrical (desalter)	-
WEIGHTS					
25	Fabricated	4.5 t	26	Empty	5.2 t
27	Operating	13.9 t	28	Hydrotest	29.5 t
NOZZLES					
Nozzle sizes per P&ID line sizing (CFU-100/200-PR-PID); manway 24" on head; PSV, vent 2", drain 2", steam-out 2", LT/LG bridle 2".					

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GENERAL					
1	Tag / quantity	D-105 / 1	2	Service	Stabiliser reflux drum
3	Orientation / support	Horizontal, 2 saddles	4	Code	ASME VIII Div.1 (U-stamp)
5	Size (ID x T/T)	ID 1.7 m x 5.5 m T/T; boot ID 0.6 m x 1.5 m	6	Volume	14.2 m3
DESIGN CONDITIONS					
7	Operating P / T	10.5 barg / 45 C	8	Design P / T	15.2 barg / 95 C
9	External design P	None (see note)	10	MDMT	-5 C
11	Corrosion allowance	3.0 mm	12	Joint efficiency	0.85
13	Hydrotest pressure	19.76 barg	14	Hold-up / residence	10 min
15	PWHT	Yes (HIC-resistant / sour)	16	Sour service	NACE MR0103
MECHANICAL					
17	Material shell / heads	SA-516-70 (HIC-tested)	18	Shell t (req / nominal)	14.2 / 16 mm
19	Head t 2:1 SE (req / nominal)	14.2 / 16 mm	20	Boot	ID 0.6 m x 1.5 m
21	Bottom of shell elevation	EL 105.000	22	Insulation	None
23	Internals	Inlet diverter, vortex breakers, weir / boot	24	Electrical (desalter)	-
WEIGHTS					
25	Fabricated	6.2 t	26	Empty	6.2 t
27	Operating	11.2 t	28	Hydrotest	20.4 t
NOZZLES					
Nozzle sizes per P&ID line sizing (CFU-100/200-PR-PID); manway 24" on head; PSV, vent 2", drain 2", steam-out 2", LT/LG bridle 2".					

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GENERAL					
1	Tag / quantity	D-106 / 1	2	Service	Splitter reflux drum
3	Orientation / support	Horizontal, 2 saddles	4	Code	ASME VIII Div.1 (U-stamp)
5	Size (ID x T/T)	ID 2.7 m x 8.5 m T/T	6	Volume	53.8 m3
DESIGN CONDITIONS					
7	Operating P / T	0.8 barg / 50 C	8	Design P / T	3.5 barg / 120 C
9	External design P	None (see note)	10	MDMT	-5 C
11	Corrosion allowance	3.0 mm	12	Joint efficiency	1.0
13	Hydrotest pressure	4.55 barg	14	Hold-up / residence	10 min
15	PWHT	No (code)	16	Sour service	-
MECHANICAL					
17	Material shell / heads	SA-516-70	18	Shell t (req / nominal)	8.2 / 10 mm
19	Head t 2:1 SE (req / nominal)	8.2 / 10 mm	20	Boot	-
21	Bottom of shell elevation	EL 106.000	22	Insulation	None
23	Internals	Inlet diverter, vortex breakers	24	Electrical (desalter)	-
WEIGHTS					
25	Fabricated	8.9 t	26	Empty	8.9 t
27	Operating	27.8 t	28	Hydrotest	62.8 t
NOZZLES					
Nozzle sizes per P&ID line sizing (CFU-100/200-PR-PID); manway 24" on head; PSV, vent 2", drain 2", steam-out 2", LT/LG bridle 2".					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		DRUMS & DESALTER DATASHEET ASME VIII Div.1 - pressure vessels D-201 - Ejector hotwell / sour water separator		DOC. No. CFU-000-ME-DS-002 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 8	
GENERAL					
1	Tag / quantity	D-201 / 1	2	Service	Ejector hotwell / sour water separator
3	Orientation / support	Horizontal, 2 saddles	4	Code	ASME VIII Div.1 (U-stamp)
5	Size (ID x T/T)	ID 1.3 m x 5.5 m T/T; boot slop oil compartment	6	Volume	7.9 m3
DESIGN CONDITIONS					
7	Operating P / T	0.05 barg / 45 C	8	Design P / T	3.5 barg / 100 C
9	External design P	None (see note)	10	MDMT	-5 C
11	Corrosion allowance	3.0 mm	12	Joint efficiency	0.85
13	Hydrotest pressure	4.55 barg	14	Hold-up / residence	15 min
15	PWHT	No (code)	16	Sour service	NACE MR0103
MECHANICAL					
17	Material shell / heads	SA-516-70	18	Shell t (req / nominal)	6.8 / 8 mm
19	Head t 2:1 SE (req / nominal)	6.8 / 8 mm	20	Boot	-
21	Bottom of shell elevation	EL 100.500	22	Insulation	None
23	Internals	Inlet diverter, vortex breakers	24	Electrical (desalter)	-
WEIGHTS					
25	Fabricated	2.5 t	26	Empty	2.5 t
27	Operating	6.4 t	28	Hydrotest	10.4 t
NOZZLES					
Nozzle sizes per P&ID line sizing (CFU-100/200-PR-PID); manway 24" on head; PSV, vent 2", drain 2", steam-out 2", LT/LG bridle 2".					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		DRUMS & DESALTER DATASHEET ASME VIII Div.1 - pressure vessels D-202 - Vacuum off-gas knock-out drum		DOC. No. CFU-000-ME-DS-002 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 9	
GENERAL					
1	Tag / quantity	D-202 / 1	2	Service	Vacuum off-gas knock-out drum
3	Orientation / support	Vertical, legs	4	Code	ASME VIII Div.1 (U-stamp)
5	Size (ID x T/T)	ID 0.6 m x 2.0 m T/T	6	Volume	0.6 m3
DESIGN CONDITIONS					
7	Operating P / T	0.05 barg / 45 C	8	Design P / T	3.5 barg / 100 C
9	External design P	None (see note)	10	MDMT	-5 C
11	Corrosion allowance	3.0 mm	12	Joint efficiency	0.85
13	Hydrotest pressure	4.55 barg	14	Hold-up / residence	10 min
15	PWHT	No (code)	16	Sour service	-
MECHANICAL					
17	Material shell / heads	SA-516-70	18	Shell t (req / nominal)	6.1 / 8 mm
19	Head t 2:1 SE (req / nominal)	6.1 / 8 mm	20	Boot	-
21	Bottom of shell elevation	EL 101.000	22	Insulation	None
23	Internals	Inlet diverter, vortex breakers	24	Electrical (desalter)	-
WEIGHTS					
25	Fabricated	0.9 t	26	Empty	0.9 t
27	Operating	1.1 t	28	Hydrotest	1.5 t
NOZZLES					
Nozzle sizes per P&ID line sizing (CFU-100/200-PR-PID); manway 24" on head; PSV, vent 2", drain 2", steam-out 2", LT/LG bridle 2".					